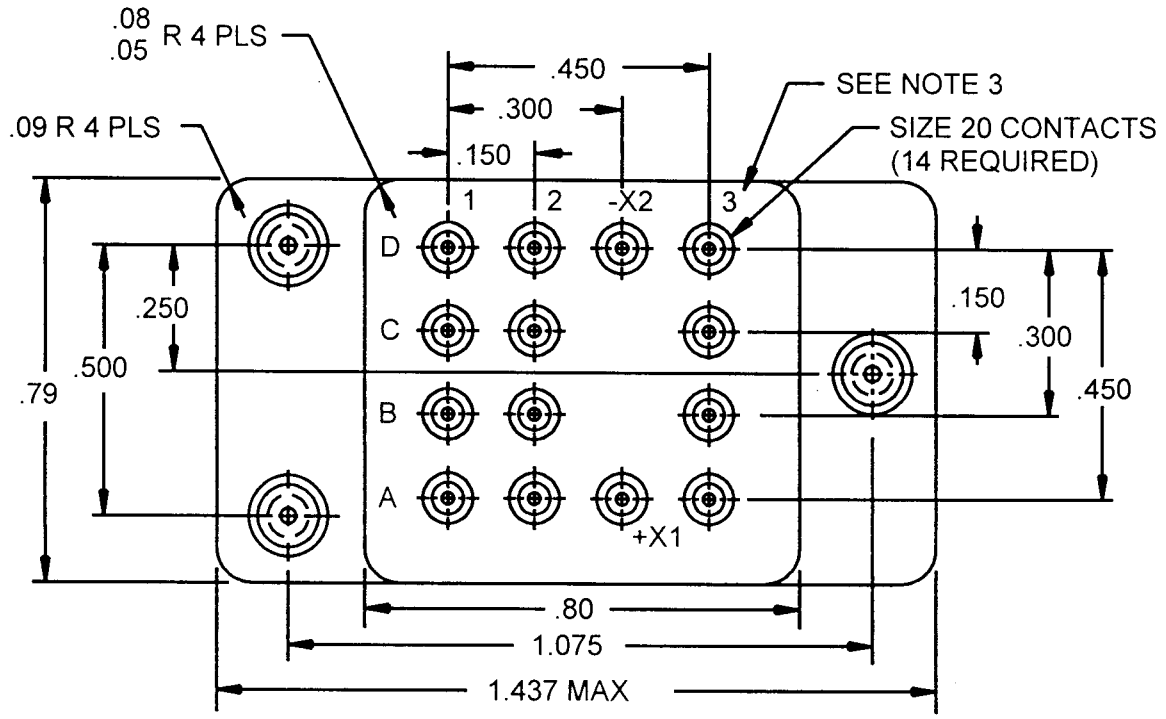
INCH-POUND
MIL-PRF-12883/44C
28 June 1996
SUPERSEDING
MIL-S-12883/44B
20 December 1989

PERFORMANCE SPECIFICATION SHEET

SOCKETS, PLUG-IN ELECTRONIC COMPONENTS, SOCKET FOR RELAYS, 4-POLE, 5 AMPERES (MIL-R-6106 AND MIL-R-83536)

This specification is approved for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-12883.

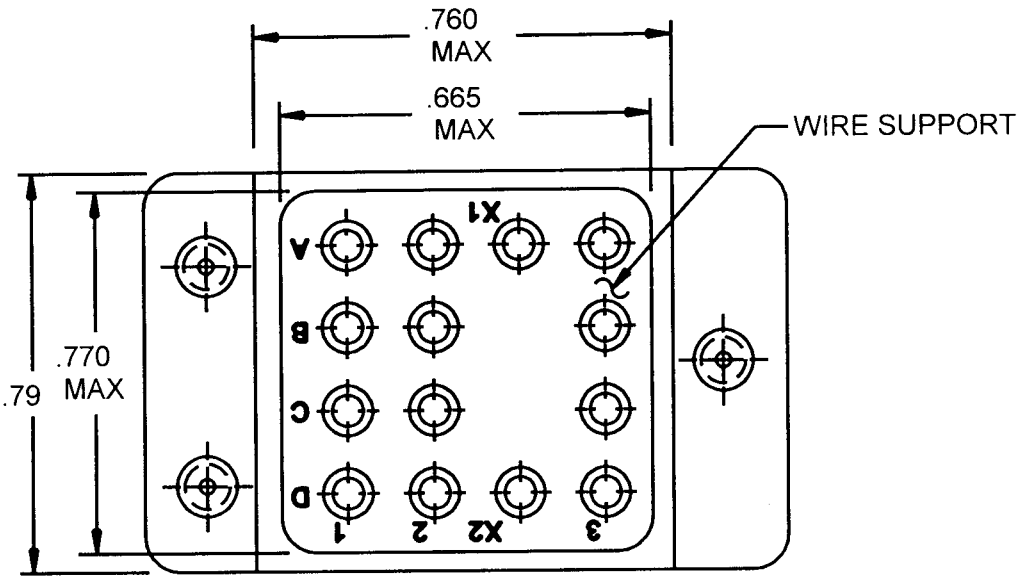


MATING FACE

FIGURE 1. Socket configuration (-01)

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

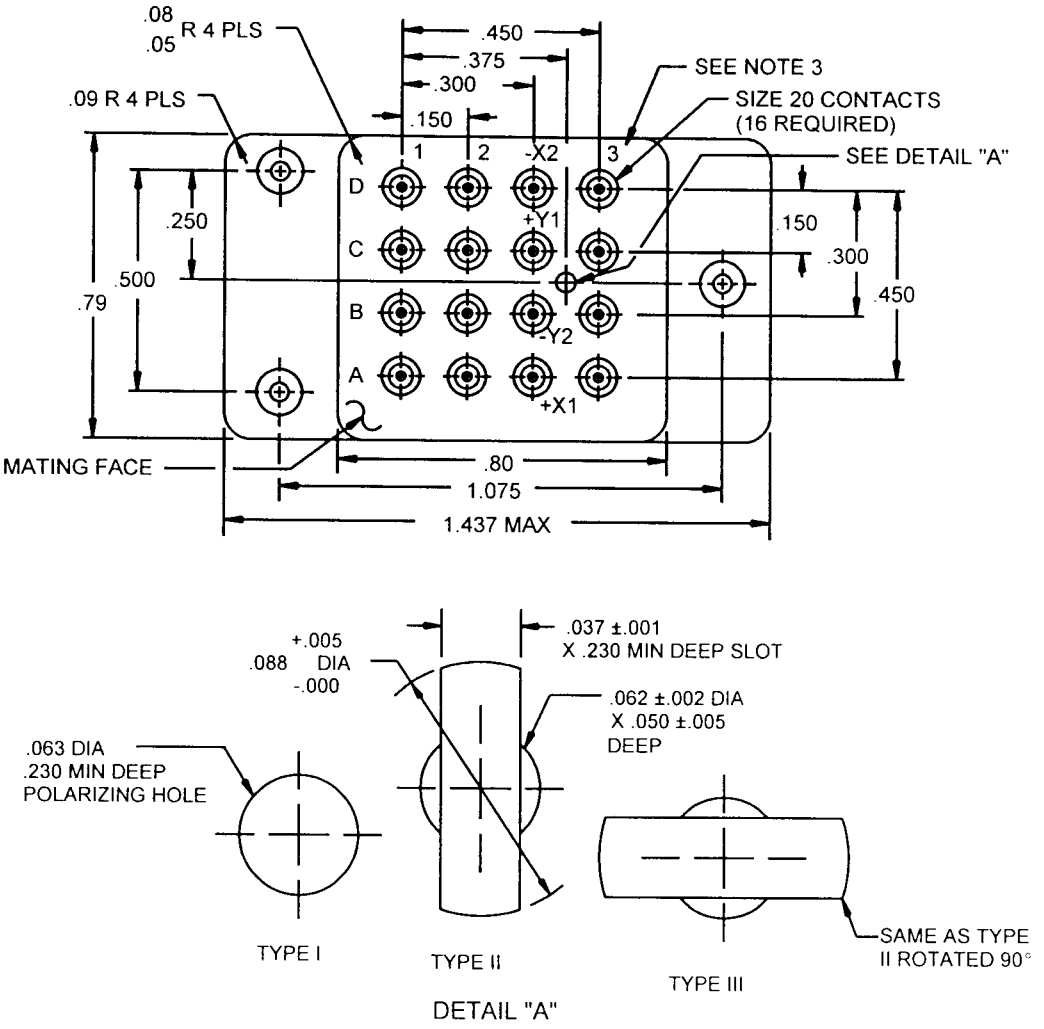
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BOTTOM VIEW

FIGURE 1. Socket configuration (-01) - Continued.

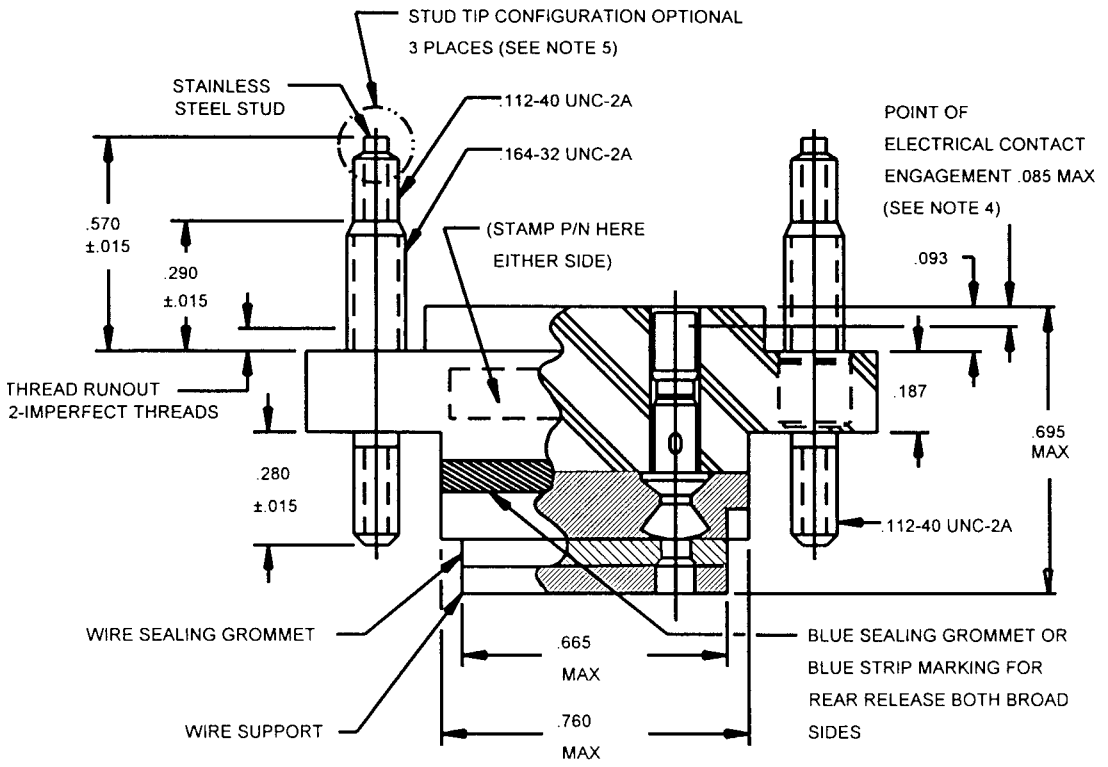
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BOTTOM VIEW

FIGURE 1. Socket configuration (-02, -03, and -04) - Continued.

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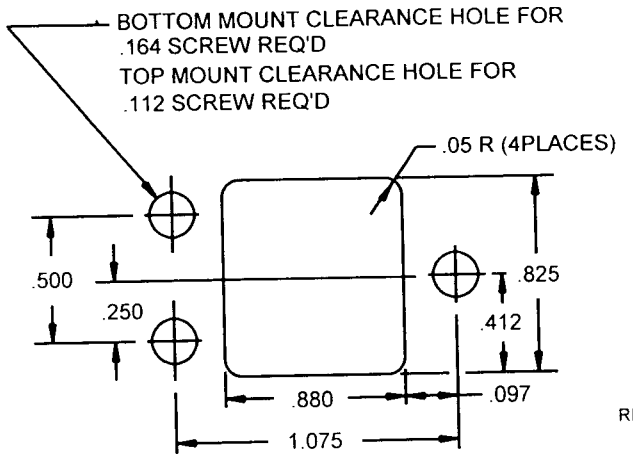


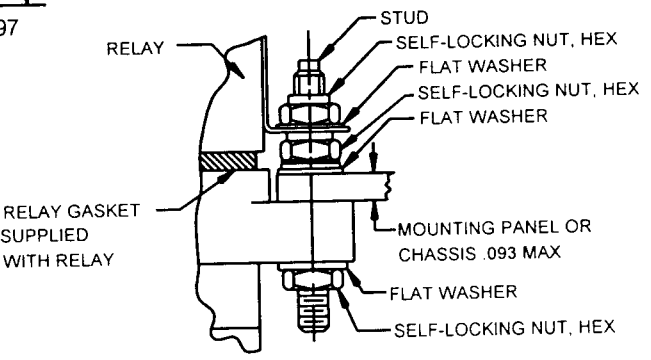
Inch	mm	Inch	mm	Inch	mm
.001	0.03	.085	2.16	.300	7.62
.002	0.05	.088	2.24	.450	11.43
.005	0.13	.09	2.3	.500	12.70
.010	0.25	.093	2.36	.570	14.48
.015	0.38	.112	2.84	.665	16.89
.035	0.89	.150	3.81	.695	17.65
.037	0.94	.164	4.17	.760	19.30
.05	1.3	.187	4.75	.770	19.59
.050	1.27	.230	5.84	.79	20.07
.062	1.57	.250	6.35	.80	20.32
.063	1.60	.280	7.11	1.075	27.31
.08	2.0	.290	7.37	1.437	36.50

NOTES
1. Dimensions are in inches.
2. Unless otherwise specified tolerances are ±.005 inch (0.13 mm) for three place decimals and ±.01 inch (0.25 mm) for two place decimals.
3. Marking shall be characters which are molded .035 inch (0.89 mm) minimum. Ink marking optional (see MIL-STD-1285).
4. Point of electrical contact engagement, from mating face of socket insulator to the socket contact.
5. The configuration of the stud tip shall be optional provided it meets all requirements of this specification.
6. Metric equivalents are given for general information only.

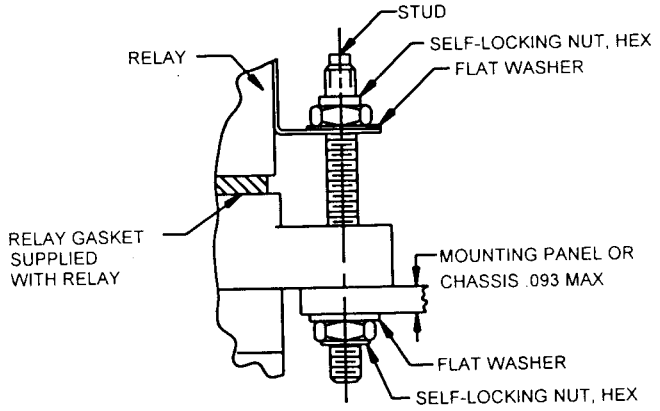
FIGURE 1. Socket configuration (-01, -02, -03, and -04 - Continued.

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BOTTOM MOUNT (TYP)



TOP MOUNT (TYP)

Inches	mm
.005	0.13
.01	0.25
.05	1.3
.093	2.36
.097	2.46
.112	2.84
.164	4.17
.250	6.35
.412	10.46
.500	12.70
.825	20.96
.880	22.35
1.075	27.31

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.01 (0.25 mm) for two place decimals.

FIGURE 2. Mounting (hardware and panel cutout).

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REQUIREMENTS:

Design and construction: See figures 1, 2, and table I.

Reference material: (see 3.1.1 of MIL-PRF-12883)

Insulator and wire support: Diallyl phthalate, in accordance with MIL-M-14, type SDG-F or any glass filled thermoplastic material in accordance with MIL-P-46179, or an approved equivalent material manufactured to a commonly accepted industry standard..

Color: Material color shall be optional providing that the color provides a contrasting background for the blue sealing grommet or blue color bands indicating rear release contacts.

Grommet: Silicon rubber in accordance with ZZ-R-765.

Mounting hardware: Corrosion resistant steel or steel with cadmium/chromate finish.

Electrical:

Test gauge: A mating relay in accordance with table I shall be used as a test gauge or mating connector.

Insulation resistance: 1000 megohms minimum.

Dielectric withstanding voltage:

Sea level: Test voltage, 1000 V rms.

High altitude (80,000 feet (24,384 meters)): Test voltage, 500 V rms.

Contact: In accordance with MIL-C-39029/101 (see table I), or approved equivalent contacts manufactured to a commonly accepted industry standard.

TABLE I. Dash numbers and characteristics.

Dash number	Socket configuration	Contact size mating end wire barrel		Number of contacts	Contact designation M39029/101	Mating relay MIL-R-83536 1/	Polarization (see figure 1 detail A)
01	-01	20	20	14	-553	/5-019, -022, -014 /6-003, -006, 011, -014, 019, -022, -029, -032	None
02	-02	20	20	16	-553	/7-022	I
03	-03	20	20	16	-553		II
04	-04	20	20	16	-553		III

1/ Reference MIL-R-83536 for supersession data on MIL-R-6106 relays.

Environmental:

Temperature range: Operating temperature range -70°C to +125°C.

Wire sealing: A resilient socket grommet is permanently bonded to the wire entry face of the socket so as to provide moisture sealing capabilities, contact M39029/101-553 wire sealing range diameter .040 min to .083 max,(1.02 to 2.11 mm), for AWG size 20 wire (for coil contacts) and AWG size 20 wire (for load contacts).

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Mechanical:

Contacts: Contacts shall be in accordance with table I. Contacts shall be crimp removable type rear release and accept relay pins size 20 (.039 through .041 inch diameter (0.99 through 1.04 mm)).

Insertion and withdrawal force: The insertion and withdrawal forces shall be as specified in table II.

TABLE II. Insertion and withdrawal forces.

M12883/44	-01	-02 through -04
Initial insertion force (max)	16 lbf (71.2 newton)	19 lbf (84.5)
Withdrawal force (min)	.61 lbf (2.7 newton)	.70 lbf (3.1 newton)
Insertion force (max), after 10 insertions and withdrawals, before vibration	18 lbf (80.0 newton)	22 lbf (97.7 newton)
Withdrawal force (min)	.52 lbf (2.3 newton)	.60 lbf (2.7 newton)
Insertion force (max), after vibration	18 lbf (80.1 newton)	22 lbf (97.9)
Withdrawal force (min)	.52 lbf (2.3 newton)	.60 lbf (2.7 newton)

Durability: Shall be in accordance with MIL-PRF-12883.

Vibration (sinusoidal): In accordance with MIL-STD-202 method 204, test condition G, or an equivalent industry standard.

a. Except that the frequency range shall be varied logarithmically between the limits of 10 and 3000 HZ.

b. Except that the procedure of method 201 of MIL-STD-202, or an equivalent industry standard, may be applied during the 10-55 Hz band of the vibration frequency range.

Vibration, (random): In accordance with MIL-STD-1344, method 2005, test condition V, letter G, or an equivalent industry standard, with a duration of 15 minutes. The mating relay shall be used as a test gage.

Shock (mechanical): In accordance with MIL-STD-202 method 213, condition C, or an equivalent industrystandard, except peak value shall be 200 g's.

Torque: Relay sockets and hardware shall be subjected to testing torque as specified in table III. Sockets shall be installed in mounting panel when test torque is applied. No physical damage will be permitted. Torque shall be maintained for a reasonable period of time to insure stud, socket, and associated hardware have not been damaged.

TABLE III. Torque requirements (installed in panel conditions).

Thread size	Torque			
	Testing		Installation	
	Inch-pounds	Newton-meters	Inch-pounds	Newton-meters
.112-40	8 + 1 - 0	0.90 + .11 -0	4 ± 1	0.45 ± .11
.164-32	20 + 1 - 0	2.26 + .1 - 0	15 ± 1	1.70 ± .11

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Socket, contact tools: See table IV, approved equivalent industry standard tools may also be used where appropriate.

TABLE IV. Contact tools.

Nomenclature	Part or Identifying Number (PIN)
Crimp tool	M22520/7-01
Positioner	M22520/7-12
Insertion and removal tool	M81969/16-01

Weight: The socket, hardware, and electrical contacts shall have a maximum weight of .055 pounds (25 grams).

Mounting hardware: The mounting hardware shall allow mounting the socket above or below the panel or chassis (see figure 2), and shall allow mounting and securing the relay to the socket without disturbing the mounted socket or access to the wiring side of the socket. The hardware shall provide the nominal .197 inch (5.00 mm) spacing between socket surface and relay mounting flange.

Mounting hardware shall be supplied with the relay socket and shall consist of the following:

Self locking nuts .112-40 (6 each) (.206 max dia x .176 max height).

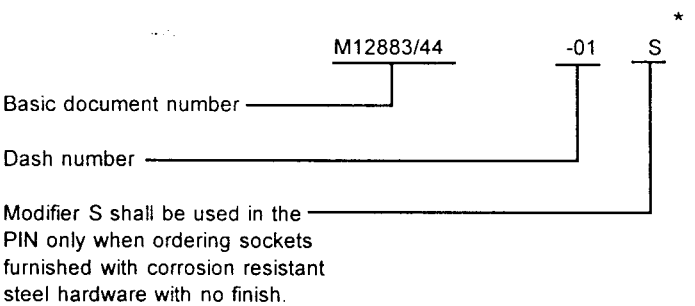
Flat washers .112 (6 each) (.224 max O.D. x .021 max thickness).

Self locking nuts .164-32 (3 each) (.290 max dia x .190 max height).

Flat washers .164 (3 each) (.290 max O.D. x .019 max thickness).

PIN: The PIN shall be marked on the socket body as shown in the example (see figure 1). The PIN shall consist of the basic number of this specification sheet and the dash number from table I.

Example



* For future acquisition of these sockets as of the effective date of revision B, 20 December 1989, parts identified with an "S" modifier shall be corrosion resisting steel (CRS), and parts without an "S" modifier shall be cadmium chromate finish. No mixing of hardware types shall be permitted.

Ordering data: Sockets without contacts may be ordered when so indicated in the ordering data (see MIL-PRF-12883). This applies only to original equipment manufacturers (OEM's) and subcontractors. All direct shipments to the government shall include all applicable contacts and mounting hardware. The PIN to be marked on the socket shall be as shown in the PIN example (see figure 1 and table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

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TABLE VI. Supersession and cross reference.

Active Government PIN	Superseded manufacturer part number	
	CAGE 58982	CAGE 99699
M12883/44-01 M12883/44-02 M12883/44-03 M12883/44-04	RSE120151 RSE120153 RSE120155 RSE120157	SE405-1020

CONCLUDING MATERIAL

Custodian:
ARMY - CR
NAVY - EC
Air Force - 85

Review activity:
Army - AR, AT, AV, ME
Navy - AS, MC, OS, SH
Air Force - 99
DLA - ES

Preparing activity:
DLA - ES

(Project 5935-4036)

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